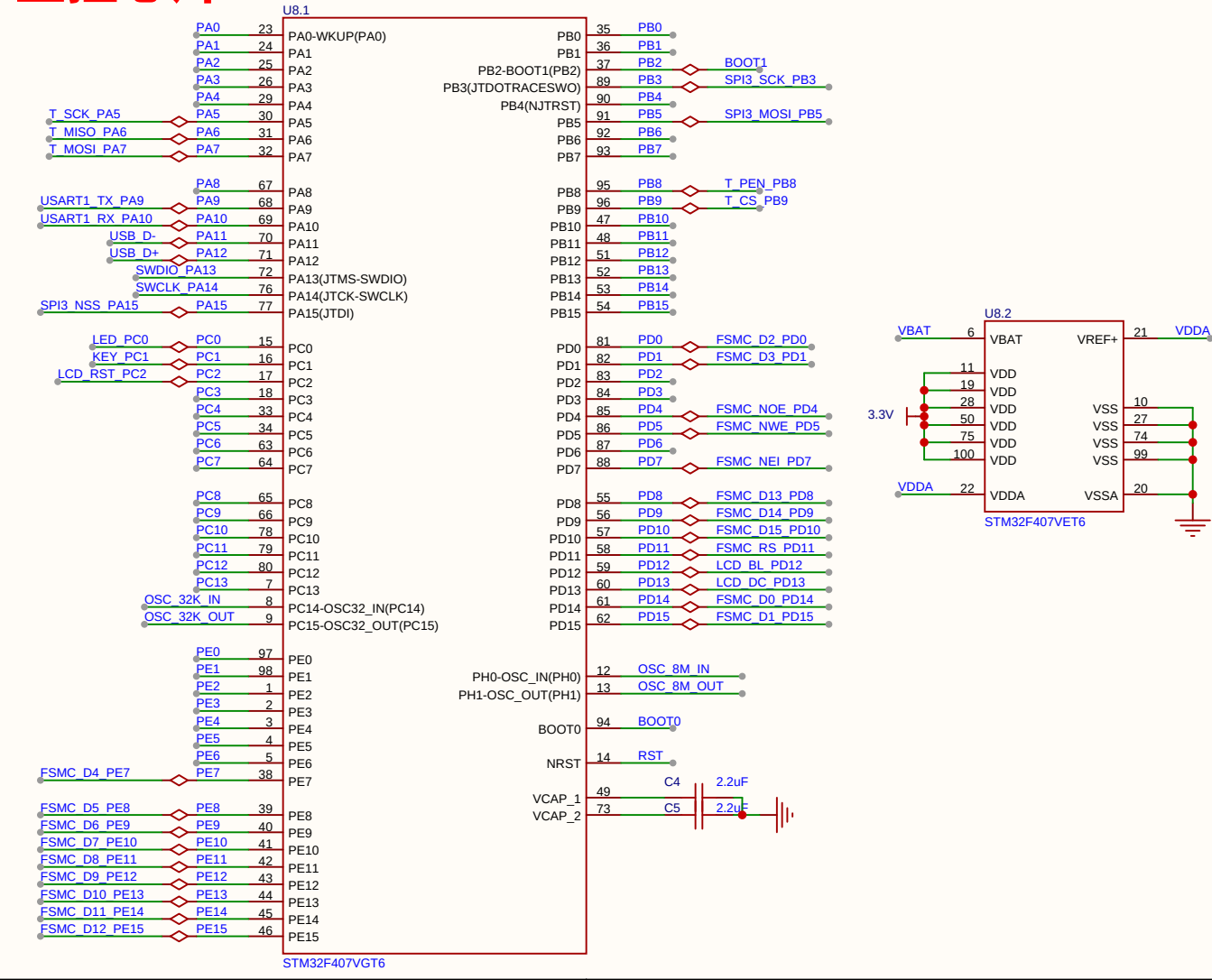
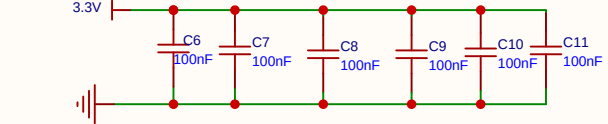


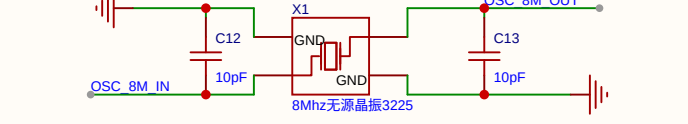
主控芯片



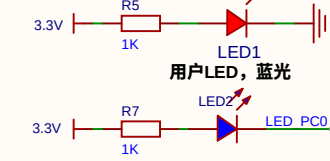
滤波电容



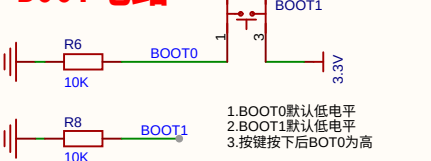
主时钟晶振



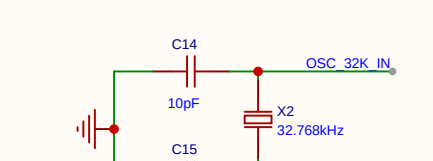
LED灯



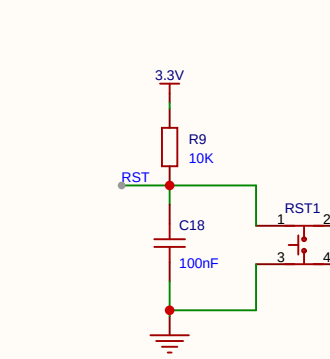
BOOT电路



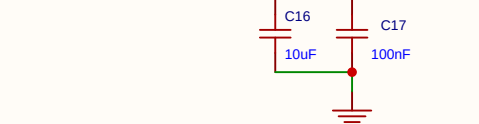
RTC时钟晶振



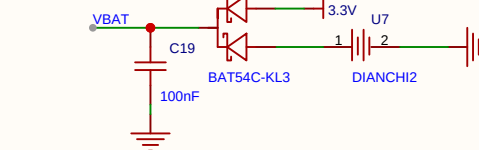
复位按键



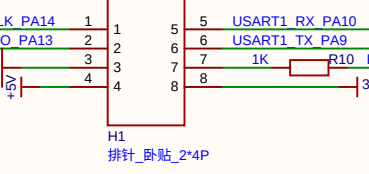
参考电压



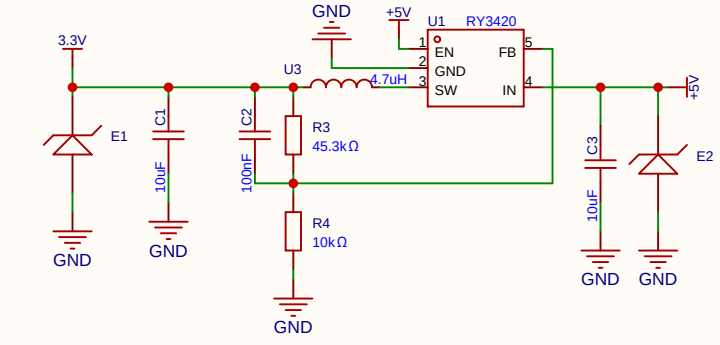
后背隐藏能源



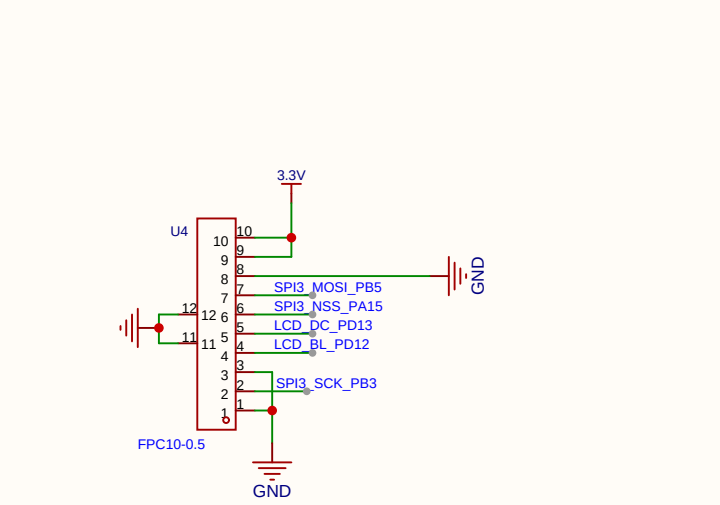
SWD和USART1接口



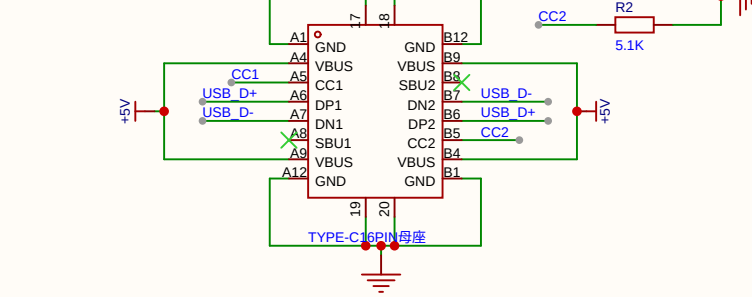
3V3电源电路



SPI液晶接口

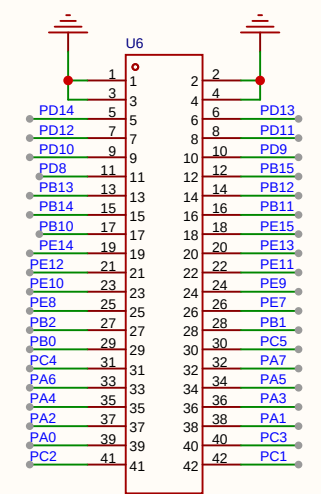
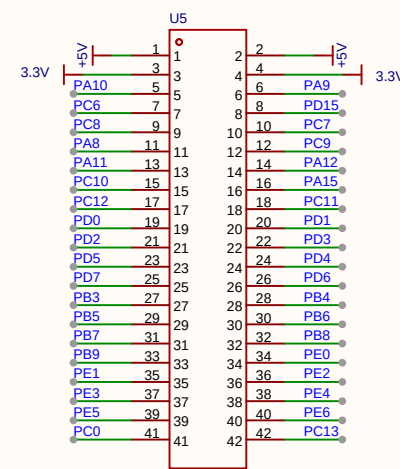
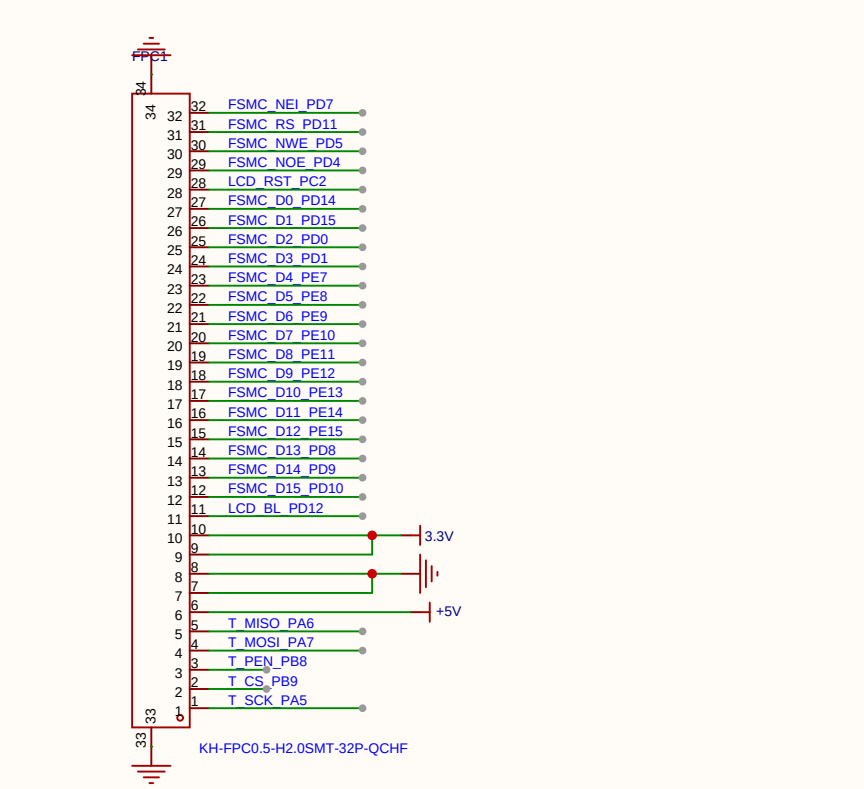


USB接口电路



1. 采用16脚的TypeC座，支持正反插
2. TypeC直接连接到STM32的PA11和PA12，可开发USB应用
3. 使用TypeC数据线连接电脑，可进行USB DFU下载用户程序

FSMC液晶接口



1. 通过2.54间距的排针引出，排针规格为2*21P
2. SWD接口和晶振占用的IO口没有引出
3. 推荐使用5V供电，再经过板载的稳压芯片得到3.3V给单片机，避免因意外输入高压而损坏单片机
4. 关于IO口的引脚复用，可以查阅STM32的数据手册，有一份完整且详细的表格供用户查